

Plaque Radiotherapy for Juxtapapillary Choroidal Melanoma

Tumor Control in 650 Consecutive Cases

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Purpose: To evaluate treatment of juxtapapillary choroidal melanoma with plaque radiotherapy and to investigate the role of supplemental transpupillary thermotherapy (TTT).

Design: Retrospective, comparative case series.

Participants: We included 650 consecutive eyes with juxtapapillary choroidal melanoma within 1 mm of the optic disc.

Methods: Eyes with juxtapapillary choroidal melanoma receiving plaque radiotherapy over a 31-year period from October 1974 to November 2005 were included in the study. The TTT and no TTT groups were analyzed separately and compared.

Main Outcome Measures: Local tumor control, metastasis, and tumor-related mortality.

Results: The median basal tumor diameter was 10 mm (range, 1.5–21) and median thickness was 3.5 mm (range, 0.5–14.8). In 481 eyes (74%), the tumor was directly adjacent to the optic disc and in 169 eyes (26%) the posterior tumor margin was between 0.1 and 1.0 mm from the optic disc. The circumpapillary extent of the tumor was <4 clock-hours in 321 eyes (50%), 4–8 clock-hours in 250 eyes (38%), and >8 clock-hours in 79 eyes (12%). Plaque radiotherapy using iodine-125 in 616 eyes (95%), cobalt-60 in 19 eyes (3%), iridium-192 in 12 eyes (2%), and ruthenium-106 in 3 eyes (<1%) delivered a median radiation dose of 8000 cGy (range, 3600–15 500) to the tumor apex and adjunctive TTT was used in 307 eyes (56%). Kaplan-Meier estimates for tumor recurrence, metastasis, and death were 14%, 11%, and 4% at 5 years and 21%, 24%, and 9% at 10 years, respectively. Eyes treated with additional TTT showed slight (statistically nonsignificant) reduction in recurrence and metastasis. Using multivariable analysis, factors predictive of tumor recurrence included foveolar tumor requiring TTT (hazard ratio, 5.07; $P < 0.001$) and greater tumor thickness (hazard ratio, 1.29 per mm increase; $P < 0.001$). Factors predictive of metastasis included greater tumor base (hazard ratio, 1.21 per mm increase; $P < 0.001$) and increasing intraocular pressure (hazard ratio, 1.11 per mmHg increase; $P = 0.020$).

Conclusions: Plaque radiotherapy for juxtapapillary melanoma provides local tumor control in approximately 80% of eyes at 10 years. In subjects who received TTT, there was slight but nonsignificant improved local tumor control and lower metastatic rate.

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The management of choroidal melanoma in close proximity to the optic disc (juxtapapillary melanoma) poses several unique challenges, owing to the posterior location of the tumor adjacent to visually significant structures. These include the choice of optimal treatment such as radioactive plaque,^{1–3} proton beam irradiation,⁴ stereotactic radiosurgery,⁵ transpupillary thermotherapy (TTT),⁶ or enucleation.⁷ If radiotherapy is chosen, there is the dilemma of whether adjunctive treatment to the tumor, such as TTT,^{6,8} should be used to ensure optimal tumor control.

It has been >10 years since our group demonstrated equivalent survival in patients with juxtapapillary melanoma treated with plaque radiotherapy versus enucleation.⁹ In prior reports, we established that plaque radiotherapy is effective for juxtapapillary melanoma encircling the optic disc (circumpapillary melanoma)¹⁰ and for juxtapapillary melanoma overhanging the optic disc.¹¹ In this study, we evaluate our overall experience of plaque radiotherapy for juxtapapillary melanoma and specifically explore the use of adjuvant TTT with regard to tumor control, metastases, and mortality.

Patients and Methods

The electronic database of patients evaluated at the Oncology Service of Wills Eye Hospital, Philadelphia, Pennsylvania, was searched for juxtapapillary choroidal melanoma treated with plaque radiotherapy. Juxtapapillary choroidal melanoma was defined as a choroidal melanoma with posterior margin touching,